LEC5 - Combinations and Permutations

Theorem 9

With m elements $\{a_1, a_2, \cdots, a_m\}$ and n elements $\{b_1, b_2, \cdots, b_n\}$, it is possible to form $m \cdot n$ pairs containing one element from each group.

Extension: If we have k sets of elements, n_1 in the first set, n_2 in the second, $\cdots$, and n_k in the k^{th} set. If you want to form a sample of one element from each k sets.

Then the number of different samples is $n_1 n_2 \cdots n_k$.

- Known as multiplication rule

Permutation: An ordered arrangement of r distinct objects is called a permutation. The number of ways of ordering n distinct objects taken r at a time is denoted by P_r^n.

Theorem 11

$$P_r^n = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}$$

Combination: The number of unordered subsets of r objects out of n is called the number of combinations. That is, the number of subsets, of size r, that can be formed from n objects. The notation is C_r^n.

Theorem 13

$$C_r^n = \binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$$

The term $\binom{n}{r}$ is generally referred to as the binomial coefficient because they occur in the binomial expansion $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$

$(x+y)^2 = x^2 + 2xy + y^2$

$1 = \binom{2}{0}$ $\quad 2 = \binom{2}{1}$ $\quad 1\binom{2}{2}$

Theorem 14

The number of ways of partitioning n distinct objects into k distinct groups containing $n_1, n_2, \cdots, n_k$ objects, respectively, where each object appears in exactly one group and $\sum_{i=1}^{k} n_i = n$ is

$$\binom{n}{n_1, n_2, \cdots, n_k} = \frac{n!}{n_1!\, n_2! \cdots n_k!}$$

The terms $\binom{n}{n_1, n_2, \cdots n_k}$ are called the multinomial coefficients because they occur in the expansion of the multinomial term $(y_1 + y_2 + \ldots + y_k)^n$.

Examples

- We have $n=3$ people (A, B, C) and $r=2$ job positions. How many ways (permutations) are there ways to assign the positions to the people?

  Thm. 12: $P_2^3 = \frac{3!}{(3-2)!} = 6 \Rightarrow \{A,B\}, \{A,C\}, \{B,A\}, \{B,C\}, \{C,A\}, \{C,B\}$

- We have $n=3$ people (A, B, C) and $r=2$ people to create a committee. How many combinations of two types of three?

  Thm. 13: $C_2^3 = \frac{3!}{2!(3-2)!} = 3 \Rightarrow \{A,B\}, \{A,C\}, \{B,C\}$

- You need to assign 9 aircrafts to 3 sorties of sizes 4, 3, and 2. How many ways can the aircrafts be assigned?

  Thm. 14: $\binom{9}{4, 3, 2} = \frac{9!}{4! \times 3! \times 2!} = 1260$

Additional Properties of Probability

$\underbrace{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$

Inclusion-exclusion formula: The Addition Law of Probability be extended to k events; $A_1, A_2, \cdots, A_k$ as

$$P\left(\bigcup_{i=1}^{k} A_i\right) = \sum_{i=1}^{k} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots + (-1)^{k-1} P(A_1 \cap \cdots \cap A_k)$$

ex.

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

choosing uniques → Without replacement
- ordered $\Rightarrow$ Permutation (P^n_r)
- unordered $\Rightarrow$ Combination (C^n_r)

choosing same object → With replacement
- ordered $\Rightarrow$ mn rule n^r
- unordered $\Rightarrow$ number of non-negative integer solutions to "$x_1 + x_2 + \cdots + x_n = r$" $= \binom{n+r-1}{n-1} = \frac{(n+r-1)!}{(n-1)!\,r!}$

Example

What are the odds of 2 people having the same birthday in a group of n people?

A: 2 people have the same birthday

A^c: No one has the same birthday

$$P(A) = 1 - P(A^c)$$
$$= 1 - \frac{365 \times 364 \times \cdots \times (365-n+1)}{365^n}$$

Person 1: $\frac{365}{365}$
Person 2: $\frac{364}{365}$
...
Person n: $\frac{365-n+1}{365}$

Example

How many unique ways can you arrange the text "abcd"

$= 4 \cdot 3 \cdot 2 \cdot 1 = 24$

4 3 2 1 choices

How many unique ways can you arrange the text "aabc"

$$= \binom{4}{2,1,1} = \frac{4!}{2! \times 1! \times 1!} = 12$$

Example

How many ways can you sort the following books while grouping subject?

- 4 stats
- 3 math
- 2 chemistry

$= 3! \cdot (4! \cdot 3! \cdot 2!)$

bundle combinations

book combinations